

The ones to remain

Tags:

Time Limit: 1 s Memory Limit: 32 MB
Submission: 342 AC: 85 Score: 96.46

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Codes

AI Code Tips

Description

There are N soldiers standing in one line. They are marked from 1 to N , from right to left. And they are given a number m . Then the soldiers numbered off, straight from the right-hand man. The one who reported a number that is the multiple of m was kept in the line. Others have to leave the line. They continue doing this till the number of people in the line is less than m . For example, if there are 10 soldiers, and $m = 3$. For the first time the soldiers who are marked 3, 6, 9 remain in the line. For the second time the soldier who is marked 9 remains in the line. Because the number of soldiers in the line is less than m , so the soldier marked 9 was the only one to remain in the line.

Now we want to know who will be the ones to remain, can you tell us ?

Input

There are several test cases in the input. Each test cases is only one line, contains two integers n and m ($3 \leq n \leq 10^9, 2 \leq m \leq n$). The input ends when $n = 0$ and $m = 0$.

Output

For each test case, output two lines. The first line contains one integer x , the number of soldiers to remain. The second line contains x integers, the numbers marked on the soldiers who remain in the line. You should output them in increasing order.

Samples

input

Copy

```
10 3
8 3
0 0
```

output

Copy

```
1
9
2
3 6
```

Source

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 15:11:45